

Dynamic Fidelity and Model Selection in Daily PM₁₀ Forecasting: A Multi-Station Forecast Evaluation Audit

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Abstract

Forecast evaluation commonly prioritises error reduction, but lower error does not necessarily imply preservation of the temporal dynamics that matter for air-quality decisions. We ask whether adding dynamic-fidelity criteria to conventional error-based statistical evaluation materially changes model eligibility in multi-step daily PM₁₀ forecasting. Using a rolling-origin benchmark of 17 Spanish monitoring stations, five model families, and seven horizons, we audit 595 station–model–horizon cells. Under Rule A (positive persistence-relative RMSE skill and significant Benjamini–Hochberg-adjusted Diebold–Mariano comparison with persistence), 277 cells are eligible. Rule B adds two operational fidelity criteria—variance retention $\alpha \geq 0.50$ and 75th-percentile exceedance recall ≥ 0.20 —and leaves 8 eligible cells. Thus, 269 of the 277 Rule-A cells change status (97.1%). Across all cells, Spearman correlation between α and skill is $\rho = -0.863$, and 101 cells meet the prespecified discordance definition: Rule A eligibility, adequate P75 recall, and $\alpha < 0.50$. These results show that error and dynamic fidelity provide distinct information for model selection in this benchmark. The thresholds are operational audit criteria, not universal standards, and the non-independent cells do not support population-level inference beyond the evaluated setting.

Keywords: PM₁₀ forecasting, model selection, dynamic fidelity, forecast evaluation, variance retention, rolling-origin audit

1. Introduction

Daily PM₁₀ forecasts support air-quality monitoring and short-term public-health decisions. Forecast verification therefore has to distinguish genuine predictive improvement from performance that is attractive only under an aggregate error score.

Conventional evaluation usually begins with a reference forecast and a scale-dependent error measure. Persistence-relative skill is useful because it asks whether a candidate improves on a simple, operationally interpretable comparator, while Diebold–Mariano testing provides a formal comparison of forecast losses [1, 2]. These tools are necessary for model comparison, but they do not describe the dynamic spread of the forecast trajectory.

A point forecast can reduce RMSE by attenuating fluctuations. Such attenuation may be acceptable for some decisions, but it can also reduce the amplitude of pollution episodes and make a forecast appear preferable under error criteria while failing a complementary fidelity check. Forecast verification literature treats accuracy, association, and other dimensions as related but non-equivalent properties [3, 4]. The practical question is consequently not whether error metrics should be discarded, but whether adding a transparent dynamic-fidelity audit changes which model cells remain eligible.

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This study asks: *Does incorporating dynamic-fidelity criteria alongside conventional error-based statistical evaluation materially alter model-selection decisions in multi-step PM_{10} forecasting?* We answer with a post-evaluation decision audit of a fixed empirical benchmark. The contributions are limited to three points:

1. an empirical audit of error–fidelity discordance;
2. a description of its variation by model family and horizon; and
3. a quantification of how an explicit fidelity requirement changes eligibility.

2. Evaluation Context and EMS Gap

Reviews of air-quality forecasting report substantial use of RMSE, MAE, correlation, and related point-error summaries when comparing statistical and machine-learning models [5, 6, 7]. These summaries are informative, but reporting of forecast variability and event sensitivity is less consistent in the audited literature.

Rolling-origin evaluation is established for time-series forecasting because it preserves temporal ordering and limits information leakage [8, 9]. Yet a leakage-free evaluation can still be decisionally one-dimensional if eligibility is determined only by error. The gap addressed here is therefore narrow: evidence is limited on whether explicitly combining error and dynamic-fidelity dimensions materially changes eligibility across sites, model families, and horizons. This is an audit of decision consequences, not a claim that the underlying metrics or the broader literature are new.

3. Decision Audit Framework

We compare two eligibility rules applied after forecast evaluation. The rules do not retrain models or create a new ranking objective; they audit how a stated requirement changes the set of eligible cells.

3.1. Error-based eligibility: Rule A

For each station–model–horizon cell,

$$\textbf{Rule A:} \quad \text{Skill} > 0 \quad \text{and} \quad \text{DM}_{BH} \text{ significant.} \quad (1)$$

Skill is persistence-relative RMSE skill. The significance flag is based on the two-sided Harvey–Leybourne–Newbold modified Diebold–Mariano comparison with Benjamini–Hochberg adjustment, as implemented in `scripts/05_dm_significance.py` [2, 10].

3.2. Fidelity-aware eligibility: Rule B

$$\textbf{Rule B:} \quad \text{Rule A} \quad \text{and} \quad \alpha \geq 0.50 \quad \text{and} \quad \text{Recall}_{P75} \geq 0.20. \quad (2)$$

The two thresholds are operational criteria chosen for this audit. They are not estimated optima, universal standards, or claims about acceptable performance in other applications. A decision change is a cell that satisfies Rule A but not Rule B.

We also define a descriptive discordant case as a Rule-A cell with $\text{Recall}_{P75} \geq 0.20$ and $\alpha < 0.50$. This isolates cells that pass the error and episode-sensitivity checks while failing the variance-retention check.

4. Empirical Benchmark

The empirical evidence is the frozen table `outputs/tables/master_diagnostic_table.csv`, whose provenance identifies 17 heterogeneous Spanish MITECO/EMEP monitoring stations and daily PM_{10} series. The canonical table contains seven horizons and five empirical model families. The source package is recovered evidence rather than a new data download; its provenance and known row-level limitations are recorded in `evidence/paper_a/README.md`.

4.1. Forecast task and models

The task is multi-step daily point forecasting evaluated at horizons $h = 1, \dots, 7$. The five model labels in the canonical evidence are direct Histogram Gradient Boosting (HGB), direct Ridge regression, fixed-order SARIMA, seasonal naive, and STL+Ridge. Persistence is the reference forecast used for skill and Diebold–Mariano comparisons; it is not counted among the five empirical model families.

The resulting unit of analysis is one station–model–horizon cell. The grid is complete: $17 \times 5 \times 7 = 595$ cells, with no duplicate keys. The cells summarise matched out-of-sample forecasts across the rolling-origin evaluation artefacts. They are not independent observations because cells share stations, time series, and model families.

4.2. Validation and preprocessing controls

The project protocol specifies rolling-origin evaluation with five expanding folds, daily frequency, a maximum horizon of seven days, persistence as the required baseline, and train-only treatment of quantities estimated within a split. P75 event thresholds are estimated from the corresponding training window. Missing verification targets are excluded rather than fabricated. Details not recoverable at the same level from the frozen aggregate are listed in `UNRESOLVED_METHOD_DETAILS.md`.

5. Metrics

5.1. Persistence-relative skill

For matched forecast–observation pairs, persistence-relative RMSE skill is

$$\text{Skill} = 1 - \frac{\text{RMSE}_{\text{model}}}{\text{RMSE}_{\text{persistence}}}. \quad (3)$$

Positive values indicate lower RMSE than persistence; they do not by themselves establish dynamic fidelity.

5.2. Variance retention

The canonical diagnostic is the predicted-to-observed variance ratio

$$\alpha = \frac{\text{Var}(\hat{y})}{\text{Var}(y)}, \quad (4)$$

with population variances ($\text{ddof} = 0$) over the matched cell pairs. Thus $\alpha = 0.50$ means that one half of the observed variance is retained; it is not a standard-deviation ratio. The canonical implementation also computes descriptive percentile-bootstrap intervals with $B = 1000$ resamples and seed 42.

5.3. Significance and event sensitivity

The Diebold–Mariano loss differential uses squared-error loss against persistence. The implementation applies the Harvey–Leybourne–Newbold small-sample correction with horizon-dependent autocovariance truncation and adjusts valid p-values using Benjamini–Hochberg within each dataset. P75 recall is the proportion of observed P75 exceedances correctly forecast as exceedances, using thresholds estimated from the corresponding training window. These diagnostics are used jointly in Rule B; none is presented as a universal replacement for error metrics.

6. Evidence and Reproducibility

All reported counts and summaries are derived from the canonical 595-row table. Its SHA-256 is recorded in the provenance manifest. The table contains exactly five model labels. Reproducible table and figure generators are included in the package, and the numerical claim-to-source map is provided in the accompanying traceability file.

The audit is intentionally descriptive. Counts are reported over a structured station–model–horizon grid, and the shared stations and time series mean that the 595 cells must not be interpreted as 595 independent replicates. Historical synthetic or quarantined material is outside the empirical evidence used here.

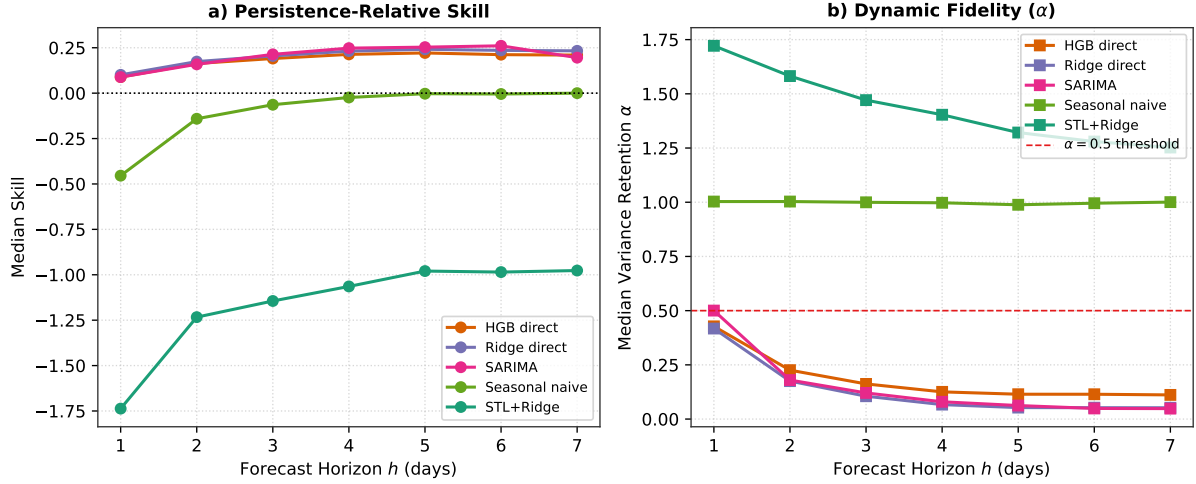


Figure 1: Median persistence-relative skill and median variance retention by horizon and model family. Each point summarises the station cells at that horizon.

7. Results

The results are presented in the order of the decision audit: error eligibility, dynamic fidelity, discordance, and the resulting change in eligibility. Table 1 reports model-level medians and counts; Table 2 reports the rule comparison.

7.1. Error-based eligibility

Rule A identifies 277 eligible cells: 111 HGB, 117 Ridge, 44 SARIMA, 5 seasonal naive, and 0 STL+Ridge. This is an eligibility count, not a claim that one model is globally best or that the cells are independent.

Table 1: Per-model diagnostic summary across 595 station-model-horizon cells ($17 \text{ stations} \times 5 \text{ models} \times 7 \text{ horizons}$). Skill is persistence-relative RMSE skill; α is variance retention; DM sig. counts BH-adjusted Diebold-Mariano cells significant at $p < 0.05$; Rule A: skill > 0 and DM significant; Rule B: Rule A and $\alpha \geq 0.5$ and $\text{recall}_{P75} \geq 0.20$; Discordant: Rule A, $\text{recall} \geq 0.20$, and $\alpha < 0.5$. Medians and means computed over all cells per model. Cells are not independent (shared stations and series).

Model	n	Median Skill	Median α	Collapse	DM sig.	Rule A	Rule B	Discordant
HGB direct	119	0.205	0.151	118/119	111/119	111/119	1/119	54/119
Ridge direct	119	0.219	0.087	118/119	117/119	117/119	1/119	43/119
SARIMA	119	0.208	0.095	110/119	44/119	44/119	1/119	4/119
Seasonal naive	119	-0.026	1.000	0/119	39/119	5/119	5/119	0/119
STL+Ridge	119	-1.107	1.399	0/119	119/119	0/119	0/119	0/119

7.2. Dynamic-fidelity behaviour

The model-level summaries show different error-fidelity profiles. Variance collapse ($\alpha < 0.50$) occurs in 118/119 HGB cells, 118/119 Ridge cells, and 110/119 SARIMA cells. Seasonal naive has 0/119 collapsed cells, while STL+Ridge has 0/119 collapsed cells but a median α above one. Figure 1 shows the corresponding horizon-wise medians. These are descriptive patterns within the benchmark, not causal or universal properties of the model classes.

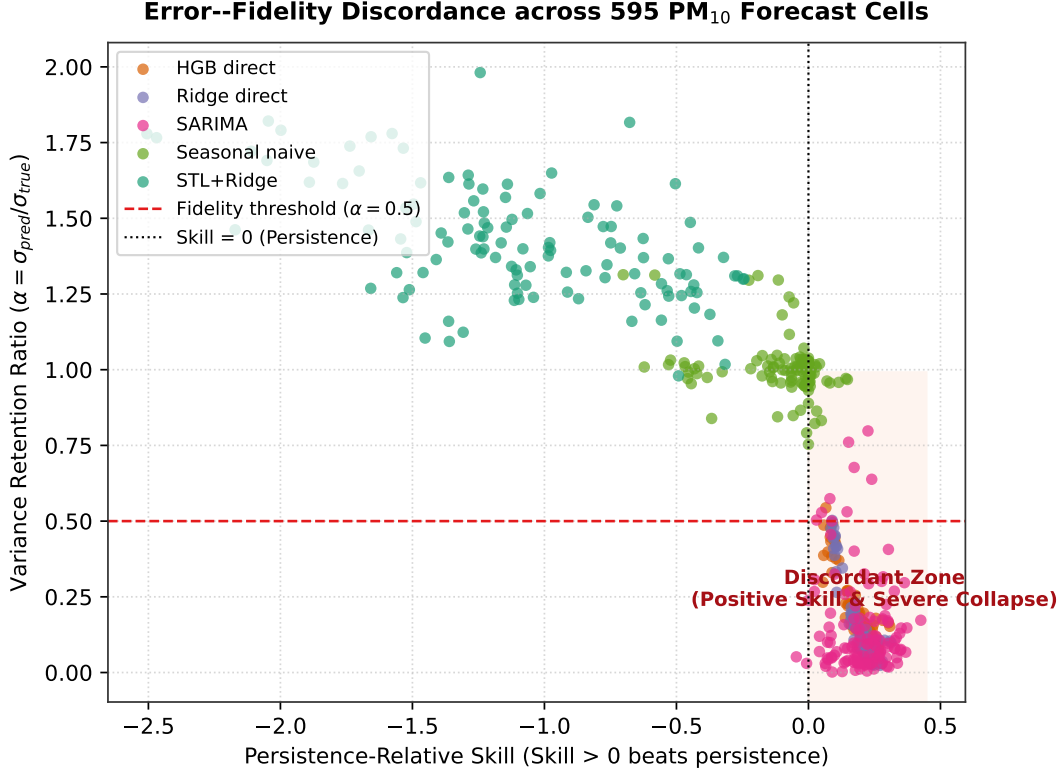


Figure 2: Persistence-relative RMSE skill versus canonical variance retention α across the 595 cells. Open points indicate Rule-A cells. The dashed line marks the Rule-B variance threshold.

7.3. Error-fidelity discordance

Figure 2 displays the joint diagnostic. Across all 595 cells, Spearman’s correlation between α and skill is $\rho = -0.863$. The correlation is descriptive and does not imply a causal mechanism.

There are 101 discordant cells under the prespecified definition (Rule A, P75 recall at least 0.20, and $\alpha < 0.50$): 54 HGB, 43 Ridge, and 4 SARIMA. Thus positive error skill and adequate P75 recall can coexist with low retained variance in this benchmark.

7.4. Effect on model eligibility

Adding the two fidelity requirements changes eligibility from 277 cells under Rule A to 8 under Rule B: HGB 111→1, Ridge 117→1, SARIMA 44→1, seasonal naive 5→5, and STL+Ridge 0→0. The 269 changed cells are therefore $269/277 = 97.1\%$ of Rule-A eligible cells. Figure 3 presents the same comparison graphically.

8. Discussion

The central result is a decision consequence: in this benchmark, 269 of 277 cells that pass the conventional error-based rule do not pass the added fidelity requirements. The audit therefore exposes a difference between being better than persistence under RMSE and retaining the observed variance of the forecast trajectory.

The discordant cells clarify the practical meaning of that difference. A model can pass the error and P75-recall screens while producing a forecast with $\alpha < 0.50$. Conversely, seasonal naive retains variance but is rarely error-eligible, while STL+Ridge avoids collapse at the cost of poor error skill. These profiles show

Table 2: Effect of adding dynamic-fidelity requirements (Rule B) to error-based eligibility (Rule A) across the 595-cell empirical benchmark. Rule A: skill > 0 and BH-adjusted DM significant; Rule B: Rule A and $\alpha \geq 0.5$ and $\text{recall}_{P75} \geq 0.20$. Decision changes indicate the number of cells eligible under Rule A but not Rule B. Cells are not independent observations.

Model	Rule A	Rule B	Changes
HGB direct	111	1	110
Ridge direct	117	1	116
SARIMA	44	1	43
Seasonal naive	5	5	0
STL+Ridge	0	0	0
Total	277	8	269 (97.1%)

why error and fidelity should be reported together when model selection is intended to support dynamic episode decisions.

The Rule-B thresholds are deliberately operational. Their purpose here is sensitivity of the eligibility decision, not prescription of a universal standard. The appropriate fidelity threshold depends on the decision, pollutant, lead time, and tolerance for missed episodes or false alarms. The result supports an auditable multi-dimensional reporting practice, not a new metric or a replacement for RMSE.

8.1. Limitations

The evidence is bounded to daily PM₁₀ from 17 Spanish monitoring stations, five model families, seven horizons, and a rolling-origin benchmark. The 595 cells share stations and underlying series and are not independent replicates. Rule-B thresholds are study-specific. The aggregate evidence is recovered and frozen; the row-level manifest notes that SARIMA skill is not independently reconstructable cell-by-cell from the distributed combined row-level file alone. The analysis uses deterministic point forecasts, does not establish causal explanations for variance collapse, and does not establish generalisation to other pollutants, countries, temporal resolutions, horizons, or model families.

9. Conclusions

In a 595-cell, multi-station daily PM₁₀ benchmark, conventional error-based eligibility identified 277 cells, while adding two explicit dynamic-fidelity criteria left 8. The 269 changed statuses amount to 97.1% of the cells eligible under Rule A, and 101 cells met the study’s descriptive discordance definition.

The methodological implication is narrow but actionable: persistence-relative error skill and dynamic fidelity answer different verification questions. Model-selection reports should make that distinction visible and state any fidelity thresholds as decision-specific conventions. These conclusions apply to the evaluated stations, models, horizons, and non-independent cell structure; they are not a universal claim about forecasting systems.

CRediT author statement

Federico García Crespi: Conceptualisation, Data curation, Formal analysis, Investigation, Methodology, Software, Visualisation, Writing – original draft, Writing – review & editing. **Julio Alberto Ramos Martínez:** Conceptualisation, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

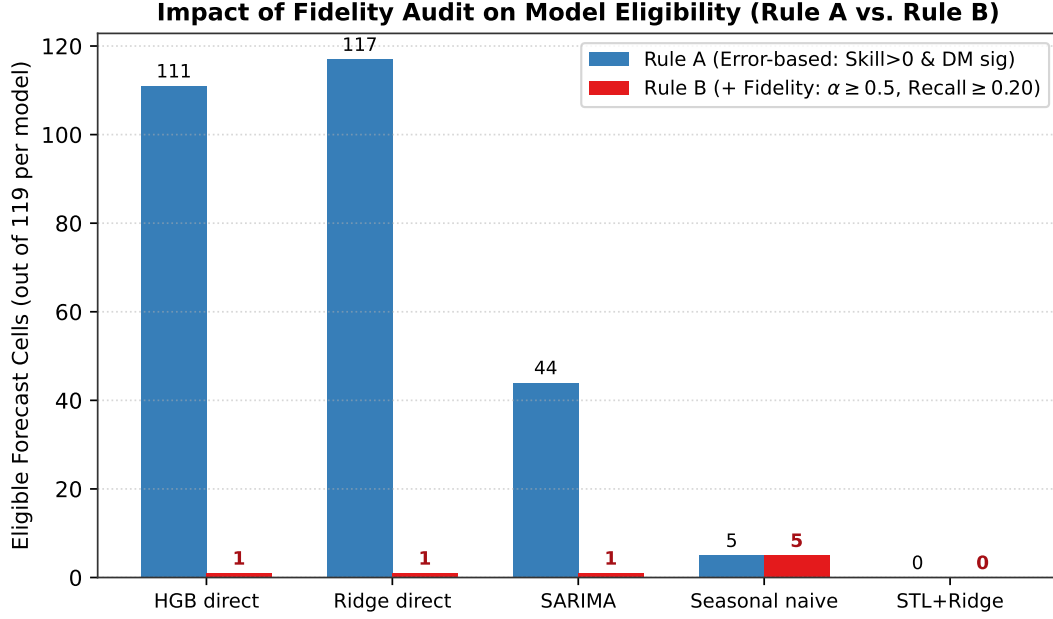


Figure 3: Eligible cells under the error-only Rule A and fidelity-aware Rule B. The total changes from 277 to 8; the 97.1% figure is the proportion of Rule-A eligible cells that fail at least one Rule-B criterion.

Data Availability Statement

The frozen aggregate evidence and provenance manifest are included in the repository. The canonical diagnostic table contains 595 station–model–horizon rows and is accompanied by SHA-256 checksums. The evidence package identifies the series as recovered Spanish MITECO/EMEP monitoring-station data; the row-level manifest and known reconstruction limitations are documented alongside it.

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Supplementary Material

S1. Reporting-audit context

Table 3: Abstract-level reporting visibility of evaluation-relevant practices in PM forecasting studies. Percentages are computed over 486 abstract-coded studies; low reporting rates indicate reporting opacity, not confirmed absence of the corresponding practice in full text.

Evaluation dimension	Explicitly reported	%	Interpretive function
Persistence or naive baseline	18/486	3.7	Minimum operational comparator
Horizon-wise reporting	11/486	2.3	Multi-step interpretability
Temporally appropriate validation	24/486	4.9	Deployment-like temporal credibility
Explicit leakage-safe safeguards	3/486	0.6	Protection against information bleed
Operational-usefulness framing	51/486	10.5	Decision relevance

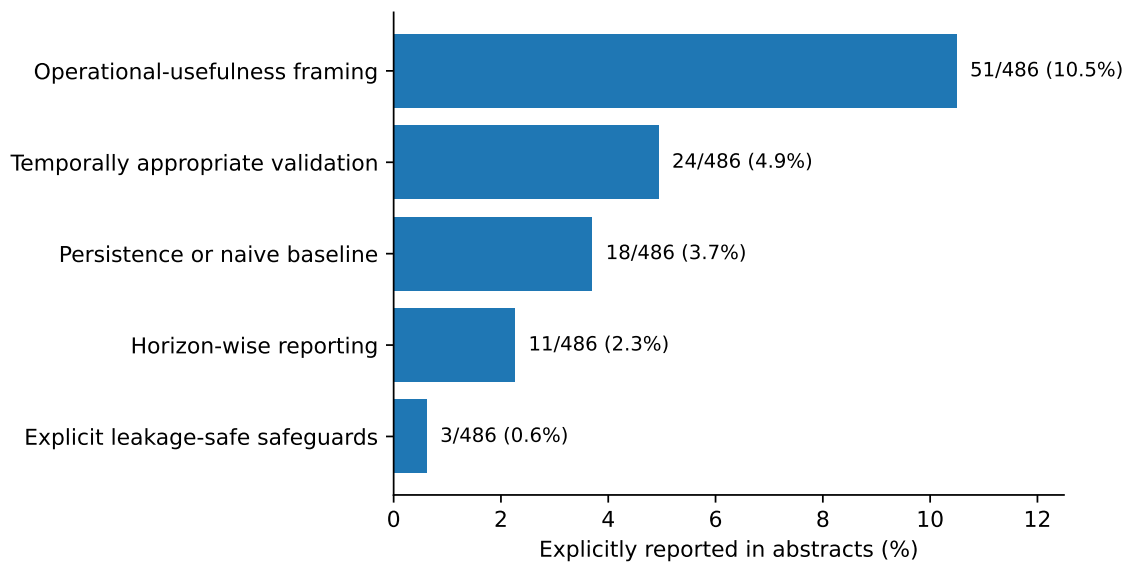


Figure 4: Abstract-level reporting visibility of key evaluation practices in particulate matter forecasting studies. Percentages are computed over 486 abstract-coded studies. Low percentages indicate reporting opacity, not confirmed absence of the corresponding practice in the full text.

S2. Monitoring stations

Table 4: Monitoring stations included in the study. Stations span industrial, traffic, urban background, suburban, and rural remote settings within the MITECO national air-quality network. Coverage refers to available daily PM₁₀ observations represented in the recovered station metadata.

Station code	Name	Area type	Altitude (m a.s.l.)	Coverage (%)
ES0446A	Elx-Agroalimentari	Suburban/Industrial	44	80.4
ES1619A	Valencia-Vivers	Urban/Residential	11	91.7
ES0012R	Zarra-EMEP	Rural Remote EMEP	855	96.0
ES1870A	Barcelona El Port Vell	Suburban/Industrial	1	98.5
ES1418A	Alagon	Suburban/Traffic	235	97.5
ES1421A	Teruel	Urban/Background	915	96.7
ES0001R	San Pablo de los Montes	Rural	923	96.1
		Remote/Background		
ES1438A	Barcelona L'Eixample	Urban/Traffic	26	95.3
ES0567A	Barcelona Zona Universitaria	Urban/Background	24	95.2
ES1417A	Huesca	Urban/Traffic	488	95.0
ES0691A	Barcelona El Poblenou	Urban/Background	3	94.9
ES0559A	Barcelona Pl. Universitat	Urban/Traffic	24	94.9
ES1856A	Barcelona Vall d'Hebron	Urban/Background	136	94.5
ES1966A	Alcaniz Capuchinos	Urban/Industrial	340	94.3
ES2011A	Sant Vicenc dels Horts	Suburban/Industrial	65	94.1
	Alaba			
ES2091A	Alcanar Montecarlo	Rural	6	93.7
		Remote/Industrial		
ES2017A	Alcanar Llar de Jubilats	Rural	7	93.6
		Near-city/Industrial		

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